

Anti-Noise Face: A Resilient Model for Face Recognition with Labeled Noise Data

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ABSTRACT

With the remarkable success of face recognition driven by large-scale datasets, noise learning has gained increasing attention due to the prevalence of noise within these datasets. While various margin-based loss functions and training strategies for label noise have been recently devised, two issues still remain to consider: 1) The explicit emphasis on specific characteristics of different types of noise is required. 2) The potential impact of noise during the early stages of training, which may lead to convergence issues, should not be ignored. In this study, we propose a comprehensive algorithm for learning with label noise. Compared to the existing noise self-correction methods, we further enhance detecting closed-set noise by introduce a closed-set noise self-correction module, and introduce a novel loss function for handling remaining noisy samples detected by an improved Gaussian Mixture Model. Additionally, we use a progressive approach, where we work through the easy examples first and then move on to the difficult ones, just as a student work through a course with the easy ones first and then the difficult ones later. Extensive experiments conducted on the synthesized noise dataset and on popular benchmarks have demonstrated the superior effectiveness of our approach over state-of-the-art alternatives.

KEYWORDS

Face recognition, Noise learning, Decision boundary, Deep learning

1 INTRODUCTION

Thanks to Convolutional Neural Networks (CNNs), face recognition has made significant progress due to the availability of large-scale training datasets. Looking through the methods based on the large-scale dataset proposed in recent years, three key factors influence the performance of CNNs in face recognition: training data, model architectures, and loss functions [28]. The margin-based loss functions [3, 15, 31] have achieved significant success. However, these methods are specifically designed for clean-face databases, which is an inevitable limitation in real-life scenarios. But factors such as poor means of acquisition, the subjectivity of annotators, and insufficient information during the annotation process will bring noise inevitably. Wang *et al.* [29] point out that the devil of face recognition lies in noise. They collate the noise in the prior popular dataset,

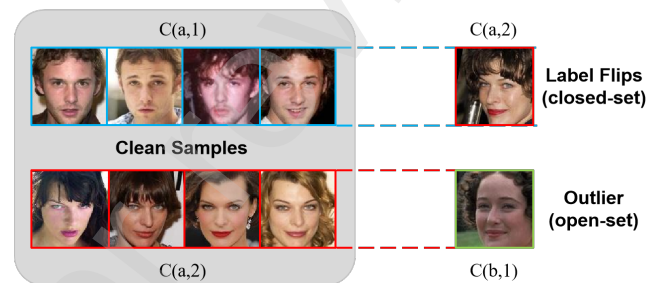


Figure 1: The illustration of deep face recognition with noisy labels in CASIA-WebFace [38]. $C(x, y)$ represents a sample of class y from database x . **Label flips** indicates the presence of mislabeled classes in the database. **Outlier** indicates that the mislabeled class belongs to a class outside of the database.

revealing that CASIA-WebFace [38], MegaFace [13] and MS-Celeb-1M [6] exhibit noise levels ranging from 9.3-13.0%, 33.7-38.3%, and 47.1-54.4%. With the increase in dataset size, the signal-to-noise ratio also increases, making noise an even more significant concern. Obtaining face datasets without noise is very expensive, as demonstrated by previous attempts to clean the IMDB-Face [29] dataset with 59K classes and 1.7M images, which the efforts of 50 annotators over a month.

Numerous studies have used noisy datasets for classification, while face recognition has several distinctive features that differ from traditional classification tasks: 1) Different distribution characteristics of data: In traditional image processing, noise is typically randomly distributed; however, in face recognition, noise may possess specific patterns or features such as occlusion, deformation, or facial expressions. 2) High robustness requirement for handling noise: Face recognition necessitates higher robustness in recognition algorithms due to the potential negative impact of noise on feature extraction and matching. Therefore, it is imperative to enhance the robustness of recognition algorithms through techniques like noise learning and denoising. 3) Differences in the noise type: Data noise can be broadly categorized into two main types: feature noise and label noise [46]. In face recognition, label noise is more prevalent and potentially more detrimental compared to other forms of noise. Label noise refers to incorrect sample labeling and can be further divided into closed-set noise and open-set noise (Figure 1). The aforementioned factors above distinguish the face noise

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problem from traditional classification tasks. However, certain common methods employed for general classification's noise processing can be adapted for use in face recognition as well, such as label cleaning [27], dataset pruning [11], sample choosing [7, 12, 16, 39], sample weighting [19, 23], and so on. Wang *et al.* [34] take the loss as the guide to reweight high-confident clean faces, making the model pay more attention to clean samples rather than noisy ones. Deng *et al.* [2] first eliminate non-dominant sub-centers and high-confident noisy data, followed by retraining the model from scratch using the automatically cleaned data. However, both approaches discard the noise and ignore the learnable characteristics of the noise. Although Hu *et al.* [9] reweight noise samples instead of discarding them, this treatment is applied uniformly to different types of noise (open-set and closed-set), failing to fully exploit the learnable information embedded in noise. Zhong *et al.* [44] assess the noise according to the distance between the label and the cosine similarity with the nearest negative class during the training process, however, due to a relatively low proportion of clean data, using dynamic data dropout may result in excessive elimination of noisy samples, causing an overemphasis on the clean data and leading to underfitting issues. Moreover, they pay little attention to how noise affects convergence issue in the early stage of training. Considering the above issues, we propose a novel training strategy that aims at learning from noisy data. Even without prior information about noise or a clean dataset for pre-training, it is still possible to train on a large-size face dataset effectively.

Our major contributions can be summarized as follows:

- We enhance the correction method for closed-set noise that is used in BoundaryFace [37], resulting in more accurate corrections of closed-set noise.
- We introduce the Early Stopping Loss (ES-Loss) mechanism to stop noise optimization and eliminate adverse impacts caused by noise.
- We propose a new noise processing framework that can efficiently process and utilize noisy data without prior knowledge present, which has been extensively tested on popular benchmarks and shown to outperform state-of-the-art alternatives.

2 RELATED WORK

2.1 Deep Face Recognition

Using margin-based Loss Functions to learn discriminative facial feature representations has become a popular approach in the field of face recognition. Margin-based Loss Function typically consists of two main components: metric-based loss function and softmax-based classification loss function. Metric-based loss including triplet loss [21], tuplet loss[24, 25], and contrastive loss [1, 5]. However, regarding the development of deep face recognition, margin softmax loss functions still have great superiority over the metric-based loss, such as no need for a complicated mining strategy. Large margin loss from softmax is a major and popular direction for face recognition [7, 15, 30, 31, 35, 36]. However, margin-based loss functions usually rely on clean datasets. Studying how models can accurately identify and learn from noise is more far-reaching than simply collecting clean data.

2.2 Deep Learning with Noisy Label

Label noise is inherent in existing face recognition datasets. Finding a noise resistance training method allows us to enlarge the scale of datasets and obtain better performance. The general idea of dealing with noise is to first model the noise, clean the data with the noise distribution, and then train the classifier with the cleaned data. Tanaka *et al.* [27] jointly optimize the model's parameters and the estimation of the noise transition matrix to deal with the noisy labels. Huang *et al.* [11] employed an extra output branch to forecast the likelihood of each input being categorized as a clean label. Subsequently, they utilized this probability score to exclude the samples with inaccurate or unreliable labels. The minimal loss selection methods [7, 12, 16, 39] leverage sample loss as a guide to select clean samples as training data for the subsequent iteration. These methods also utilize prior knowledge such as the noise level, a small clean set, and a pre-trained model to aid in the selection process. Other researchers [19, 23] have proposed a sample re-weighting strategy that assigns varying degrees of importance to different types of samples. By doing so, the model can give different levels of attention to each sample, depending on their significance for the learning process.

2.3 Face Recognition with Noisy Label

There are two primary approaches to handling noise in training datasets: direct discarding and noise re-weighting. Zhong *et al.* [45] propose a noise-robust loss functions company with a training strategy that treats the head data and the tail data unequally. Wu *et al.* [41] raise a practical automatic label noise cleansing framework for face recognition based on two cascaded graph convolutional networks. Wang *et al.* [33] incorporate a reliable vector that guides the feature learning process to reduce intra-class variation, which is learned during training and is utilized to represent the common features of each class. Zhong *et al.* [44] dynamically adjust the data dropout rate during training by computing the distance difference between each sample and other samples within the same class to evaluate the reliability of each sample and its contribution to the model. Wang *et al.* [34] use two networks that swap samples to reduce the accumulation of errors and take the loss as a cue to synthesize the noisy samples. Deng *et al.* [2] design sub-centers for each class to encourage one dominant sub-class that contains the majority of clean faces, as well as non-dominant sub-classes that relax the intra-class constraint to improve the robustness of label noise. The methods above identify the noise samples and throw them away. Ding *et al.* [4] demonstrate that discarding noise samples facilitates the establishment of semi-supervision. Nevertheless, Hu *et al.* [9] re-weight the clean samples which were separated by the Gaussian mixture model. Apart from dealing with the noise samples with the same strategy, Wu *et al.* [37] specifically address closed-set noise and propose to determine closed-set noise from the perspective of decision boundaries and correct it. Besides noise learning, Kim *et al.* [14] adaptively adjust the learning rate and regularization during training, along with a novel loss function that aligns and normalizes facial landmarks, to reduce interpersonal variations.

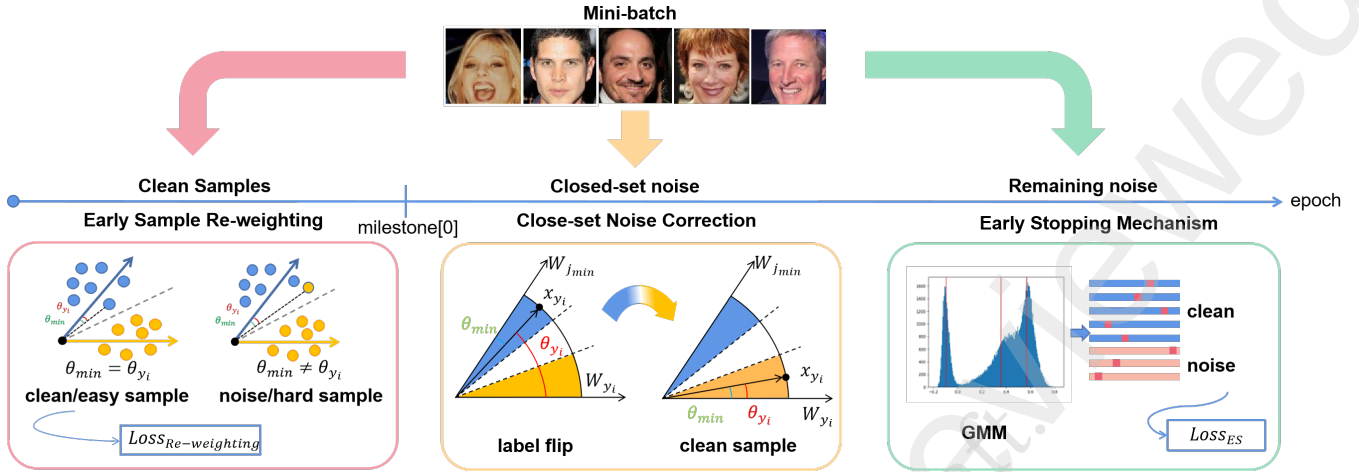


Figure 2: The overview of our method, including three modules: clean samples re-weighting, closed-set noise label correction, and noise early stopping mechanism. θ_{y_i} : The angle between x_i and the labeled identity W_{y_i} , θ_{min} : The smallest value of angles between x_i and targets $W_j \in \{W_1, W_2, \dots, W_C\}$. In the Close-set Noise Correction module, the identified closed-set noise samples (marked in blue) are transformed into clean samples (marked in yellow) after correction.

3 METHOD

Our objective is to learn as much information as possible from the noisy samples in case of the noise ratio is unknown. We aim to accurately identify and correct closed-set noise, while minimizing its impact on model performance. To achieve this, we propose a novel strategy for closed-set noise correction and introduce a new loss function for handling remaining noise. The framework (Figure 2) consists of three modules: 1) Noise label self-correction: We detect and correct the noisy samples in the training dataset to effectively exploit the closed-set noise, and transform the noise into simple samples that can be learned by model. Compared to the previous self-correction method [37], better results can be obtained by using the decision boundary measured by cosine similarity to detect closed set noise. 2) Early stopping mechanism: We employ an improved Gaussian mixture model to generate the threshold value for identifying noise, and then introduce a new loss function that halts optimization of features associated with noise samples at an early stage. 3) Progressive Learning: To prevent negative effects of noise during initial training stages on model’s performance, we propose prioritizing relatively simple samples before the first change in learning rate occurs. This ensures that the model pays more attention to the easy samples, thereby reducing the impact of noise during the initial training phase.

3.1 Noise Label Self-correction

Label noise is ubiquitous in face recognition tasks, where data annotation is often carried out manually or automatically with error-prone algorithms. To mitigate the negative impact of label noise, we identify and correct closed-set noise from the perspective of decision boundaries. Margin-based softmax loss function [3, 15, 31] by setting the margin to achieve Intra-class aggregation and inter-class separation. Binary-classes scenarios with classes c_1 and c_2 are usually used to explain the decision boundary between two classes.

w_1 and w_2 denote weight vectors for c_1 and c_2 . In this paper, we consider using the label class (C_{y_i}) and the nearest negative class ($C_{j_{min}}$) of the sample instead of c_1 and c_2 to determine whether the sample is a closed-set noise. As discovered by Wu *et al.* [37], closed-set noise is likely to be distributed within the decision boundary of the nearest negative class. Therefore, using the negative class closest to the sample as the correction result is reasonable. Apart from the Arcface [3] decision boundary used in BoundaryFace [37], we conducted experiments with other classical margin-based decision boundaries and found a more suitable one for correcting the closed-set noise in face recognition. Let θ_{y_i} denote the angle between the sample and the label class, and θ_{min} denote the angle between the sample and the nearest negative class.

Normalized-Softmax: Normalize-Softmax improves the softmax loss by normalizing weight vectors W , the decision boundary for determining whether the sample labels are correct or closed-set noise is:

$$\cos(\theta_{y_i}) = \cos(\theta_{min}) \quad (1)$$

It is not quite robust to noise because there is no decision margin: any small perturbation around the decision boundary can change the judgment of closed set noise. Across multiple classes, the similar distances between adjacent classes may result in probabilistic interactions between some classes.

SphereCorrection: SphereFace improves the softmax loss by introducing an extra margin, the decision boundary of SphereCorrection for determining whether the sample labels are correct or closed-set noise is:

$$\begin{aligned} C_{y_i} : \cos(m\theta_{y_i}) &\geq \cos(\theta_{min}) \\ C_{j_{min}} : \cos(m\theta_{min}) &\geq \cos(\theta_{y_i}) \end{aligned} \quad (2)$$

However, The decision boundary of the SphereFace method is relatively fuzzy, because its decision boundary has a certain class of large distances. This can lead to probabilistic interactions between boundary classes, which can affect recognition performance.

ArcCorrection: ArcFace adds a margin to the angle to improve softmax, and ArcCorrection's decision boundary is as follows:

$$\begin{aligned} C_{y_i} : \cos(\theta_{y_i} + m) &\geq \cos(\theta_{min}) \\ C_{j_{min}} : \cos(\theta_{min} + m) &\geq \cos(\theta_{y_i}) \end{aligned} \quad (3)$$

In this case, each category corresponds to a unit circle in a high-dimensional space, which may cause the distribution of categories on the decision boundary to be relatively fuzzy, affecting the accuracy of recognition results. In addition, each class in the Additive angular margin computations a fixed angular shift matrix, which can be costly. The CosFace method has no similar operation and is much faster.

CosCorrection: CosFace defines a decisive margin in cosine space rather than the angle space, the decision boundary of CosCorrection is given by:

$$\begin{aligned} C_{y_i} : \cos(\theta_{y_i}) &\geq \cos(\theta_{min}) + m \\ C_{j_{min}} : \cos(\theta_{min}) &\geq \cos(\theta_{y_i}) + m \end{aligned} \quad (4)$$

$\cos(\theta_j)$ is maximized while $\cos(\theta_{y_i})$ is minimized for $\cos(\theta_{y_i})$ (similarly for C_{y_i}) to perform the large-margin classification. By limiting the cosine similarity between the same class, the CosFace method ensures that the set of images of the same class is more compact, thus making the decision boundary more clear. From the distribution of the cosine of the angle, a small perturbation around the decision boundary (dashed line) is less likely leads to an incorrect decision.

BoundaryFace [37] had made a breakthrough in closed-set noise correction. However, we find that using cosine similarity instead of angle as the metric to judge whether a sample needs to be corrected, can more accurately find the closed-set noise existing in the training set. Margin is used to control the interval range between different classes. If it is too small, it may cause mutual interference between feature vectors of different classes; if it is too large, it may cause the classification boundary to be too strict and affect the classification accuracy. The difference between the decision boundaries of ArcFace and Cosface is whether the margin is added on the Angle or the cosine similarity. Specifically, Angle measurement typically uses Euler angles or rotation matrices to compare the difference in rotation Angle between two face images. Even if the face features in the two images are the same, if there are factors such as rotation and tilt, this measurement method cannot accurately express the similarity between the two faces. Cosine similarity is a more direct and accurate measure. It can represent two face images in vector form, and then calculate the cosine value of the Angle between them, the smaller the Angle, the higher the similarity. ArcFace is used as the benchmark for face recognition models in the industry because of its faster convergence speed and better geometric properties, but for the task of distinguishing closed set noise from clean samples, Cosface's decision boundary is more applicable. Specifically, comparing the distance relationship between the cosine similarity corresponding to the label and the cosine similarity of the nearest negative class, if the distance between $\cos \theta_{y_i}$ and $\cos \theta_{min}$ exceeds the margin m , the label of that sample will change from y_i to j_{min} .

3.2 Early Stopping Mechanism

Two main problems need to be addressed for noise handling: 1) How to identify noise from a large number of training samples; 2) How to deal with or exploit noise effectively. However, these two steps are often intertwined, as there is no guarantee that the samples identified as noise by the model are actually noise, the samples rectified by the correction module are actually closed-set noise. Therefore, simple dropout operations are not recommended. Inspired by early stopping strategies, we integrated the handling of noisy samples into the model training process. Early stopping originally refers to stopping training when the loss on the test set no longer decreases. Similarly, in dealing with noisy samples, we stop optimizing them as soon as they are identified. On this account, we use a Gaussian Mixture Model (GMM) for identification. As for the noise samples identified by the GMM, setting their cosine similarity with respect to all classes to 1 means that the model considers that no further optimization is necessary for the samples. The above process is implemented by setting a new loss function.

GMM is one of the most widely used unsupervised modeling techniques, which assumes that the data is a mixture distribution composed of multiple Gaussian distributions. The goal of GMM is to estimate the parameters of each Gaussian distribution to best fit the original data distribution. The noise and clean samples studied in this paper belong to two different distributions, so they can be separated from the overall data set by a Gaussian mixture model. We can regard mixture models as generalizing k -means clustering to incorporate information about the covariance structure of the data as well as the centers of the latent Gaussians. The probability density function of the Gaussian mixture model of K components is defined as:

$$p(x | \theta) = \sum_{k=1}^K \alpha_k \phi(x | \theta_k) \quad (5)$$

where α_k denotes the probability that the sample belongs to the K -th components, $\phi(x | \theta_k)$ denotes the probability density function of the K -th Gaussian distribution. We use the Expectation-Maximization (EM) procedure to fit the GMM.

Based on the $\cos \theta_{y_i}$ histogram distributions, we can estimate the probability that a sample is mislabeled. To model the distribution of $\cos \theta_{y_i}$, we fit a Gaussian mixture, which reveals that the randomly generated model parameters have different distributions for different types of samples, and belong to different Gaussian distributions. In the Gaussian Mixture Model (GMM), parameter K represents the number of Gaussian distributions or components. Selecting the correct K value is crucial as it directly affects the model's complexity and accuracy. After a period of training, the histogram changes from one Gaussian distribution peak to two. As Hu *et al.* [9] proved that the left peak represents noisy samples, while the right peak represents clean samples. However, we found that when setting $K = 2$, although the data distribution can be roughly fitted, the second Gaussian distribution deviated from the true peak. While taking $K = 3$ as the condition of GMM gives us not only a better fitting result but also a good indicator to recognize whether the sample is a noise. It can be seen from figure 3 of the comparison results. From a logical perspective to explain the experiment result, in the case of setting $K = 2$ for GMM and the presence of hard samples, the hard samples may be located in the overlapping

region of the two Gaussian distributions or on the boundary region between the two Gaussian distributions. This is because hard samples are usually located at the overlapping position of the two distributions, making it difficult to determine which distribution is more suitable for these data points. In such cases, increasing the number of Gaussian distributions can be attempted to better adapt to the complexity of the data. As for the threshold of recognizing whether the sample is noise, it is most appropriate to take the mean value of μ_1 and μ_3 as the comparison standard of cosine similarity with the sample label class. Samples that $\cos \theta_{y_i} < d$ are considered to be noise.

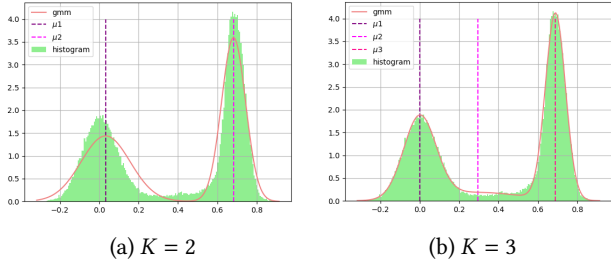


Figure 3: The fitting results of Gaussian mixture model under different K .

For the samples that meet the condition of $\cos \theta_{y_i} < d$, we first tried to discard noise samples directly, but the experimental results were not satisfactory (Table 3). Output samples recognized as closed set noise by the self-correcting module and samples recognized as noise by GMM. We found that it was unreasonable to correct closed set noise and discard residual noise at the same time. Perhaps some noise was discarded before it was corrected, and some noise discarded might be clean samples. Therefore, in the case of closed set noise correction, the best strategy to deal with the remaining noise is to stop the optimization of it in advance, rather than discard it. Specially, we increase the noisy samples' $\cos \theta_{y_i}$ to 1. Here, $\cos \theta_{y_i}$ represents the distance between a sample and its label class center feature, and $\cos \theta_{y_i} = 1$ indicates that the sample is perfectly close to its class center. Furthermore, CNN only tries to force other class centers away from it. On the one hand, the proposed method reduces the effect of noise. On the other hand, it also minimizes inter-class similarity and maximizes intraclass similarity. Margin-based softmax loss function are typically defined as follow:

$$L = -\frac{1}{N} \sum_{i=1}^n \log \frac{e^{s \cdot T(\cos \theta_{y_i})}}{e^{s \cdot T(\cos \theta_{y_i})} + \sum_{j=1, j \neq y_i}^c e^{s \cdot \cos \theta_j}} \quad (6)$$

We make improvements in the margin-based loss function to realize the above idea, then we have:

$$T(\cos \theta_{y_i}) = \begin{cases} 1, & \cos \theta_{y_i} < d \\ \cos(\theta_{y_i} + m), & \cos \theta_{y_i} \geq d \end{cases} \quad (7)$$

To test the module's performance in detecting open-set noise and closed-set noise, 20% of the samples in the original CASIA-WebFace [38] dataset are label reversed, and the other 20% are replaced with data from the MageFace [13] dataset, which are used as training

data to record the recognition accuracy of the two types of noise. It further suggests that our method can precisely recognize noise samples, and it is more efficient to recognize open-set noise than closed-set noise.

3.3 Progressive Learning

At the beginning of training, the model has not established any basic knowledge base for facial features, just like a student who knows nothing about the course he is about to learn. If all images are treated equally during the entire training process, the face images with different characteristics cannot be fully utilized. Before more complex face image recognition, we consider learning images that are relatively simple for the model, such as the images with sufficient lighting conditions, no occlusion, and clear pixels. Projection into the learning process is to focus on learning relatively simple parts first, and then learn complex knowledge with a part of the foundation, which can better resist noise and other image changes, thus improving the overall recognition accuracy of the model. The learning rate changes during the training process also follow this rule, first coarse-grained learning at a faster speed, and then slowly slow down. In addition, because the simple face image contains less feature information, the model can be trained and recognized faster, and the recognition speed can be improved.

Three important questions arise: How to define a simple face? How to treat simple faces? At what stage is such an operation implemented? Here are the answers in turn: For the definition of 'simple face', we find that for images with clear pixels and without any occlusion, the label identity (θ_{y_i}) and model prediction (θ_{min}) can reach agreement at the early stage of training, while for relatively difficult samples, it takes longer training time to achieve high accuracy. However, the noise, which contains both open set noise and closed set noise, fails to obtain accurate results throughout the training process. Figure 4 illustrates this conclusion.

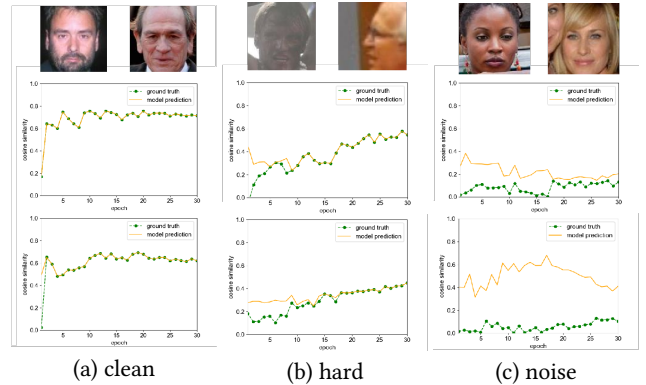


Figure 4: The comparison of the label identity of clean, hard, and noisy samples and the degree of fluctuation of the model prediction during training.

Therefore, this characteristic is used as the basis for determining clean samples in this paper. However, researchers often use weighted samples to adjust the contribution degree of samples to

the CNN model [9, 34, 44]. Weighting is particularly useful for emphasizing the importance of easy or difficult samples. We achieve this by implementing the weighting approach from the paper [34].

As mentioned above, in the same way that 'a student takes a course', it is easier to learn something simple first, so focus on learning simple faces at the beginning of training. With the deepening of learning, the knowledge base of the model grows, and more and more samples are identified as simple and clean samples. The specific training strategy is: before the learning rate is changed, the model will pay more attention to the samples whose class is the closest to the sample and the class corresponding to the sample label.

4 EXPERIMENTS

4.1 Datasets

Training Datasets. We trained our model using CASIA-WebFace and a synthesized noise dataset based on it. Previous research [31] has shown that the CASIA-WebFace database contains 9.3%-13.0% noisy samples, with 7.7% of images being labeled as noise based on manual annotation. For the synthesized data, we added both open-set and closed-set noise to the original dataset. Open-set noise was randomly selected from MegaFace [13], while closed-set noise involved flipping the label of original samples.

We conducted experiments on datasets with varying proportions of noise. Specifically, on the one hand, to test the effectiveness of our closed-set noise correction method, we added different ratios of closed-set noise to the original dataset and verified the robustness of the model in handling different scales of closed-set noise. On the other hand, to evaluate the performance of our framework, we added different proportions of open-set and closed-set noise to the original dataset.

Testing Datasets. We utilized six benchmark datasets to evaluate our model's performance, including LFW (Labeled Faces in the Wild) [10], CALFW (Cross-Age Labeled Faces in the Wild) [43], CPLFW (Cross-Pose Labeled Faces in the Wild) [42], AgeDB-30 (Age Database) [17], CFP-FP [22], and RFW (Racial Faces in the Wild) [32]. The LFW dataset contains over 13,000 face images of 5,749 individuals, collected from the internet, with varying poses, lighting, expressions, and backgrounds. CALFW extends LFW by including images of people of different ages, ranging from 16 to 62 years old. CPLFW focuses on cross-pose face recognition, including poses ranging from frontal to profile views. AgeDB-30 contains images of people with ages ranging from 2 to 62 years old, captured under various lighting conditions and facial expressions. CFP-FP emphasizes large pose variation in unconstrained settings. Finally, RFW is commonly used in cross-racial face recognition research, containing images from four different racial groups: Asian, Black, Indian, and White.

4.2 Experiment Settings

For a fair comparison, we follow the approach of ArcFace [3] to crop the faces with five landmarks [26, 40] to a size of 112×112 , and we use the ResNet50 [8] network as our CNN architecture. For data preprocessing, we normalize each pixel in the images (which range between [0,255]) by subtracting 127.5 and dividing by 128. We also horizontally flip all the training faces with a probability of

0.5 for data augmentation. The model is trained using the stochastic gradient descent (SGD) algorithm on CASIA-WebFace, with a batch size of 64. We set the momentum to 0.9 and the weight decay to 0.0005. Our framework is implemented in PyTorch [18]. Regarding the hyper-parameters in the margin-based method, we set the scale s to 32 and the margin m to 0.3. We conducted experiments with different margins in various noise environments and found that due to the presence of noise, it is necessary to increase the constraint on the inter-class margin.

4.3 Experiments on Synthesized Data

To begin with, we use the ResNet-50 [8] model trained with ArcFace on the CASIA-WebFace [38] database as our baseline model. The effectiveness results of the above three optimization modules are verified as follows.

Effectiveness of Closed-set Noise Correction. Closed-set noise has a more significant impact on the model than open-set noise [29]. The most effective use of closed-set noise is when the labels of closed-set noise can be accurately identified and corrected, which can directly expand the dataset by including these samples as clean samples. Currently, in face recognition, the optimal method for dealing with closed-set noise is BoundaryFace [37]. Setting aside the algorithm part of difficult sample mining, the main idea is to judge the noise based on the decision boundary of ArcFace [3]. To examine whether the decision boundary is the best for judging closed-set noise, we apply the decision boundary of other classical face recognition algorithms, such as SphereFace [15] and Cosface [31] to judge closed-set noise by inverting 20% of the samples' label in the original CASIA-WebFace dataset. Before starting the correction of closed-set noise, pre-training is conducted using the respective decision boundaries before the first milestone. As shown in Table 1, from a horizontal comparison perspective, the closed-set noise correction based on the decision boundary is effective, which greatly improves the robustness of learning in noisy environments. From a longitudinal comparison, the decision boundary based on CosFace for correcting closed-set noise has the best performance. Therefore, we determined the strategy to use the CosCorrection method as the closed-set noise correction module. At the same time, we investigate the feasibility of the method on datasets with different closed-set noise rates. The experimental results in Table 2 show that our method has obvious advantages. To more intuitively highlight the advantages of our method, we compared the accuracy rate and recall rate of the three decision boundary correction methods for closed-set noise in the training process. It can be seen from Figure 5 that the correction strategies based on CosFace and ArcFace performed better and had better performance when the correction strategies were set for the first time after pre-training. However, the correction strategy based on SphereFace slowly showed its correction ability in the later stage. The reason is that the hyper-parameter λ of the weight decay factor is difficult to control, which leads to poor robustness of the model.

Effectiveness of Early Stopping Loss (ES-Loss). In addition to the closed-set noise, it is inevitable that classes outside of the database will appear due to poor data collection methods. For this type of data, we propose using an improved Gaussian mixture model to identify and set a new loss function for optimizing the

Table 1: The performance (%) of closed-set noise correction methods based on different decision boundaries, where RFW (Afr), RFW (Asi), RFW (Cau), and RFW (Ind) denote the African, Asian, Caucasian, and Indian test protocols in RFW, respectively.

Method	LFW	CALFW	CPLFW	SLLFW	AgeDB	CFP-FP	RFW(Asi)	RFW(Cau)	RFW(Ind)	RFW(Afr)
SphereFace	98.88	89.77	84.75	94.38	89.92	92.14	78.80	87.45	84.05	78.08
SphereCorrection	98.95	90.80	86.60	95.62	90.87	93.00	82.77	89.85	86.38	81.02
ArcFace	98.87	92.28	88.10	96.93	91.98	92.38	84.53	91.33	88.35	84.45
ArcCorrection	99.23	92.88	87.65	97.45	93.80	93.33	86.03	93.02	90.08	86.60
CosFace	99.25	92.88	87.98	97.21	92.90	93.21	85.13	92.35	89.12	85.03
CosCorrection	99.27	93.47	88.33	98.07	94.50	93.83	86.98	94.25	90.50	87.37

Table 2: The performance (%) of closed-set noise correction methods with different closed-set noise ratios (%), where RFW (Afr), RFW (Asi), RFW (Cau), and RFW (Ind) denote the African, Asian, Caucasian, and Indian test protocols in RFW, respectively.

ratio(%)	Method	LFW	CALFW	CPLFW	SLLFW	AgeDB	CFP-FP	RFW(Asi)	RFW(Cau)	RFW(Ind)	RFW(Afr)
10	BoudaryF1	99.4	93.18	89.32	97.85	94.02	94.3	86.5	93.32	90.33	86.95
	CosCorrection	99.43	93.75	89.67	98.3	94.93	94.84	86.9	94.07	90.42	88.5
20	BoudaryF1	99.23	92.88	87.65	97.45	93.8	93.33	86.03	93.02	90.08	86.6
	CosCorrection	99.27	93.47	88.33	98.07	94.5	93.83	86.98	94.25	90.5	87.37
30	BoudaryF1	99.2	92.27	83.22	96.97	92.72	88.26	84.88	91.97	88.72	84.68
	CosCorrection	99.18	93.02	86.33	97.33	93.33	91.90	85.52	92.92	89.33	86.43

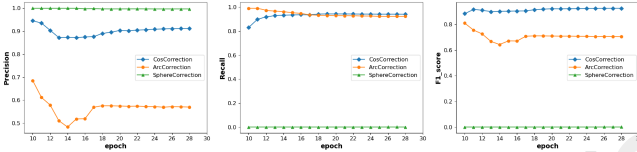


Figure 5: Precision, Recall, and F1 score of our close-set noisy detection strategy in different training epochs based on different decision boundaries.

Table 3: The Performance of different treatments on noise samples with 20% open-set noise and 20% closed-set noise.

Method	LFW	AgeDB	CFP-FP
ArcFace	98.87	89.55	89.56
Discard	98.60	89.25	90.70
ES-Loss	99.17	92.75	93.03

model. Under the guidance of our proposed improved Gaussian mixture model and a new loss function for the identified noise, our model has achieved high performance. The results presented in Table 3 demonstrate that stopping the optimization for noise is more effective than discarding the noise samples.

To evaluate the module’s ability to identify different types of noise, we conducted an experiment to measure the precision rate of noise recognition with only the ES-Loss guidance. We divided our synthesized datasets into three parts: synthesized label flips, synthesized outliers, and original samples, denoted as F , O , and A , respectively. The original samples consist of clean samples and a small amount of unknown noise. Samples recognized as noise are labeled as F' , O' , and A' . By using the indices of the recognized noise samples in the training stage, we can calculate the rate of correctly identified noise that comes from actual synthesized noise.

The results presented in Table 5 indicate that it is more effective to recognize outliers than label flips. However, when combined with the closed-set noise correction module, the label noise problem can be well resolved. Since the details of the noise are known, using the indices of the samples recognized as noise in the training stage, we can calculate the recognition rate of actual synthesized noise. As shown in Table 5, this further indicates that it is more effective to identify outliers than label flips. Combined with the closed-set noise correction module, the problem of label noise can be effectively resolved.

Effectiveness of Progressive Learning. On a dataset with 40% noise (*o10c30* means 10% of open-set noise and 30% of closed-set noise, similarly for other symbols.), we plotted the proportion of clean/easy samples and noise/hard samples in the dataset in the training progress, as shown in Figure 6. A sample is defined as *clean* if its ground truth class center and its nearest negative class center are uniform, otherwise, it is labeled as *noise*. The figure highlights the crucial role played by the relationship between a sample’s ground truth class center and its nearest negative class center in optimizing face recognition models. Taking the relationship as a clue to judge whether the sample is a clean one or a noisy one is reasonable. In addition, taking inspiration from the way noise is handled in general classification, implementing sample re-weighting in the training process can be of guiding significance. This involves first learning the *easy* features as a basis, thereby weakening the impact of the *bad* features. Specifically, in the early stages of training, giving more attention to the samples that appear to be clean at first glance can help to reduce the influence of noise on the model’s discrimination ability to some extent. As shown in Table 4, the last line of the ablation results shows that the performance is improved after adding the early sample re-weighting module.

Effectiveness of Anti-noise Face. By combining the modules mentioned above, a comprehensive noise processing method is created. The ablation study is presented in Table 4. Although Rolnick

Table 4: The experiment result of ablation study on a training dataset with 20% open-set noise and 20% closed-set noise, where RFW (Afr), RFW (Asi), RFW (Cau), and RFW (Ind) denote the African, Asian, Caucasian, and Indian test protocols in RFW, respectively.

Method	LFW	CALFW	CPLFW	SLLFW	AgeDB	CFP-FP	RFW(Asi)	RFW(Cau)	RFW(Ind)	RFW(Afr)
ArcFace	98.87	90.43	84.4	94.87	89.55	89.56	80.97	88.37	85.83	79.72
ES-Loss	98.95	92.37	87.58	97.03	92.57	92.96	85.18	91.28	88.35	84.10
CosCorrection	99.00	92.48	90.21	92.20	84.73	96.60	84.42	91.35	87.98	84.15
Our	99.17	92.37	87.07	97.02	92.75	93.03	84.33	91.40	88.13	84.22

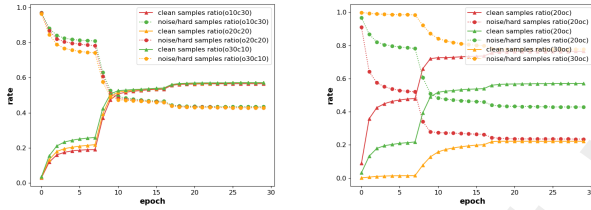
Table 5: The accuracy of noise recognition by the improved GMM model when training with 20% open-set noise and 20% closed-set noise.

Epochs	$O'/O(\%)$	$F'/F(\%)$	$A'/A(\%)$
12	99.99	71.87	11.80
14	99.99	85.73	14.68
16	99.99	85.96	13.42
18	99.99	81.01	10.54
20	99.99	79.94	9.73
22	99.99	79.58	9.55

of noise, it is necessary to increase the constraint on the inter-class margin.

Table 6: The performance (%) of our method with different hyper-parameter m on datasets that contain different noise mixing ratios (%).

C(%)	O(%)	m	LFW	AgeDB	CFP-FP
10	30	0.3	99.13	92.68	93.20
		0.5	99.08	92.30	92.49
30	10	0.3	90.00	92.33	92.79
		0.5	99.10	93.22	92.97



(a) 40% noise in training dataset (b) open-set:closed-set=1:1

Figure 6: (a): ratio variation of clean/noisy samples during training with the same overall noise ratio but different ratios of open and closed set noise. (b): ratio variation of clean/noisy samples during training with the same ratio of open and closed set noise but different overall noise rates.

et al. [20] have proved that deep CNNs can still perform well on noisy datasets when their scale is large enough, indicating that CNNs inherently possess robustness against noise. However, when noise accounts for a substantial proportion of the training data, CNNs become ineffective in preventing overfitting. Therefore, the training process involves the following steps: in the early stage of training, we add the clean faces re-weighting module based on the algorithm implemented in CosFace [31], which incorporates intra-class aggregation and inter-class distance. After ten epochs of training, the model has a certain level of recognition ability for data with noise, and the decision boundary of CosCorrection is used to determine whether the sample is closed-set noise. If the sample falls within the decision boundaries of other negative classes, its label is corrected to the nearest negative class. Since our improved Gaussian mixture model can effectively identify open-set noise, we use the ES-loss to handle the remaining open-set noise. Regarding the margin in CosFace [31], Table 6 shows that due to the presence

In summary, we conducted experiments on a dataset that includes both open-set and closed-set noise (the dataset construction method is the same as previously described). The experimental results are shown in Table 7, which demonstrate that the noise processing strategy proposed in this paper has strong robustness and comprehensiveness, and can effectively handle most noise situations. Based on the training data set containing 40% noise (open-set: closed-set = 1:1), the average accuracy of the 10 test sets is taken as the measurement standard, and the experimental results of different methods are plotted, as shown in Figure 7.

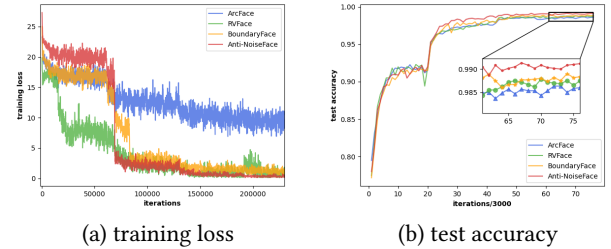


Figure 7: The training loss and test accuracy of corresponding methods under the training set with 40% noisy samples (open-set:closed-set = 1:1).

5 CONCLUSION

In this paper, we propose a comprehensive face recognition method to address label noise by employing different processing strategies based on the characteristics of open-set and closed-set noise. For closed-set noise, we introduce an improved correction strategy, while employing an improved Gaussian mixture model to detect open-set noise. Moreover, a novel loss function is designed to terminate the optimization of open-set noise at an early stage, which

Table 7: The performance (%) of different loss functions when the training set contains different mixing noise ratios, where C denotes closed-set noise ratio, O denotes open-set noise ratio. RFW (Afr), RFW (Asi), RFW (Cau), and RFW (Ind) denote the African, Asian, Caucasian, and Indian test protocols in RFW, respectively.

C(%)	O(%)	Method	LFW	CALFW	CPLFW	SLLFW	AgeDB	CFP-FP	RFW(Asi)	RFW(Cau)	RFW(Ind)	RFW(Afr)
10	30	ArcFace	98.78	91.48	86.12	96.42	91.67	92.17	82.95	90.30	87.85	82.72
		RVFace	98.93	91.73	86.35	96.37	91.95	91.49	83.08	90.52	87.65	82.82
		BoundaryFace	99.17	91.95	86.22	96.55	92.32	92.40	84.15	91.05	88.17	83.28
		Ours	99.13	92.58	87.92	97.05	92.68	93.20	84.95	91.67	89.10	85.25
20	20	ArcFace	98.60	90.72	83.98	95.27	89.33	89.03	81.10	87.95	85.58	79.58
		RVFace	98.78	91.03	84.38	94.98	90.57	90.01	81.43	88.12	85.53	80.05
		BoudaryFace	99.20	92.32	87.20	96.97	92.68	93.28	84.78	91.88	88.8	84.48
		Ours	99.17	92.37	87.86	97.02	92.75	93.84	84.33	91.40	89.24	85.20
30	10	ArcFace	98.30	89.62	82.03	93.52	87.98	87.64	79.85	86.42	84.20	77.22
		RVFace	98.02	89.85	82.58	93.45	87.82	86.97	79.63	86.40	83.96	77.25
		BoudaryFace	99.10	92.50	82.22	96.88	93.38	88.24	83.77	91.18	88.18	83.67
		Ours	99.10	92.63	87.63	97.10	93.21	92.97	84.10	91.88	88.77	84.65

proves more effective than direct discarding. Additionally, we incorporate re-weighting measures for clean samples to address convergence issues caused by noise. Through extensive experiments on popular test datasets, our approach demonstrates superior generalization and effectiveness compared to state-of-the-art methods.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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